Exercise 19.2

1. State Newton's Laws of motion and explain how from the first law we obtain a definition of the force and from the second a measure of force.

Sol^n: ***Newton's Law's of Motion***

First law : Every body continues in its state of rest or of uniform motion in a straight line unless compelled by some external force to change that state.

Second law : The rate of change of momentum is directly proportional to the impressed force and takes place in the direction of applied force.

Third law : To every action, there is an equal and opposite reaction. The first law gives the definition of the force and second law gives the measurement of the force and third law specifies the property of force.

First law gives definition of force

Newton's first law serves to define a force. It states that the state of rest or of uniform motion of a body will change only when some external force is applied on it. An object on the table does not move unless it is pushed or pulled. Hence force is defined as that which change or tends to change the state of rest or uniform motion of a body in a straight line. Thus, the external agency is needed which change the state of rest of uniform motion of a body. This agency is called force.

Second law gives measurement of force

Suppose a force F acts on a body of mass m for a time t and changes its velocity from u to v.

Impressed force $\propto$ Rate of change of its momentum

i.e. $F \propto \frac{mv - mu}{t}$ or, $F \propto m\frac{(v - u)}{t}$

But $\frac{v - u}{t} = \frac{\text{change of velocity}}{\text{time}}$ = acceleration = a(say)

$\therefore$ $F \propto ma$

$\therefore$ $F = kma$, where k is proportionality constant.

This equation enable us to measure of force.

2. How large a force is required to bring a 1000 kg car moving with a velocity of 100 ms^{-1} to rest at (a) a distance of 1000 m; (b) in 20 second.

Sol^n: mass of the car (m) = 1000 kg

initial velocity (u) = 100 ms^{-1}

final velocity (v) = 0

force (F) = ?

a) Now, $v^2 = u^2 + 2as$

$\Rightarrow$ $0^2 = (100)^2 + 2 \cdot a \cdot 1000$ $\Rightarrow 2000\ a = -\ 10000$ $\Rightarrow a = -5\ ms^{-2}$

$\therefore$ $F = ma = 1000 \times 5 = 5000$ N.

b) Now, $v = u + at$ $\Rightarrow 0 = 100 + a \cdot 20$ $\Rightarrow a = -5\ ms^{-2}$

$\therefore$ $F = ma = 1000 \times 5 = 5000$ N.

3. A constant force of 10 N acting on an object reduces its velocity from 15 ms^{-1} to 5 ms^{-1} in 2s. Find the mass of the object.

Sol^n: force (F) = 10 N

initial velocity (v) = 15 ms^{-1}

final velocity (v) = 5 ms^{-1}

time taken (t) = 2 sec

mass of the object (m) = ?

Now, acceleration $= \frac{v - u}{t} = \frac{5 - 15}{2} = -5\ ms^{-2}$

The negative sign shows the retardation

$\therefore$ $F = m \times a \Rightarrow 10 = m \times 5$ $\Rightarrow m = 2$ kg.

4. A 12.0 g bullet is accelerated from rest to a speed of 700 ms^{-1} as it travels 20 cm in the barrel of a gun. Assuming the acceleration to be uniform, find how large the accelerating force was ?

Sol^n: mass of the bullet (m) = 12.0 g = 0.012 kg

initial velocity (u) = 0

final velocity (v) = 700 ms^{-1}

distance covered (s) = 20 cm = 0.2 m

accelerating force (F) = ?

Now, $v^2 = u^2 + 2as$

$\Rightarrow$ $(700)^2 = 0^2 + 2 \times a \times 0.2$ $\Rightarrow 490000 = 0.4a$ $\therefore a = \frac{490000}{0.4} = 1225000\ ms^{-2}$

Hence, the required accelerating force, $F = m \times a = 0.012 \times 1225000$ N = 14700 N.

5. A bullet moving at 250 ms^{-1} penetrates 5 cm into a tree trunk before coming to rest. Assuming that the force exerted by the tree trunk is uniform, find its magnitude. Mass of the bullet is 10 g.

Sol^n: velocity of the bullet (u) = 250 ms^{-1}

mass of the bullet (m) = 10 g = 0.01 kg

initial velocity (u) = 250 ms^{-1}

final velocity (v) = 0 ms^{-1}

Let a is the retardation, then

$v^2 = u^2 - 2as$

$\Rightarrow$ $0^2 = (250)^2 - 2 \times a \times 0.05$ $\Rightarrow 0.1a = (250)^2 \Rightarrow a = 625000\ ms^{-2}$

$\therefore$ $F = ma = 0.01 \times 625000$ N = 6250 N.

6. A force of 25 newtons acts on a mass of 0.50 kg starting from rest. Find

a) the acceleration in ms^{-2}

Sol^n: force acting (F) = 25 N

mass of the body (m) = 0.50 kg

initial velocity (u) = 0

Now, $F = m \times a$

$\Rightarrow$ $a = \frac{F}{m} = \frac{25}{0.50} = 50\ ms^{-2}$.

b) the final velocity after 20 s.

Sol^{n}: Let, v be the velocity after 20 seconds.

$\therefore \quad v = u + at = 0 + 50 \times 20 = 1000\ ms^{-1}$.

c) the distance moved in 20 s.

Sol^{n}: Let s is the distance moved in 20 secs.

$\therefore \quad s = ut + \frac{1}{2}at^2 = 0 + \frac{1}{2} \times 50 \times (20)^2 = 10000$ m.

7. A train of mass 327 tones moves at the rate of 108 kmh^{-1}, after the steam is shut off, it is brought to rest by the brakes in 50 m. Find the force exerted, assuming it to be uniform and assuming g = 9.81 ms^{-2}.

Sol^{n}: mass of the train (m) = 327 tones

initial velocity (u) = 108 $kmh^{-1} = \frac{108 \times 1000}{60 \times 60} = 30\ ms^{-1}$

final velocity (v) = 0

distance described (s) = 50 m

force exerted (F) = ?

Let, a be the retardation due to brakes, then

$v^2 = u^2 - 2as$

$\Rightarrow \quad 0^2 = (30)^2 - 2 \,.\, a \,.\, 50 \qquad \Rightarrow \quad a = \frac{900}{100} = 9\ ms^{-1}$

$\therefore \quad F = \text{mass} \times \text{acceleration} = 327 \times 9 \text{ tones } ms^{-2} = \frac{327 \times 9}{9.81}$ tones wt. = 300 tones wt.

8. On turning a corner, a motorist rushing at 36 km/hr finds a child 51 m ahead, he stops the car within one metre of the child by the application of the brakes. Calculate the retarding force and the time required to stop the car. The total mass of the car and the passenger = 2000 kg.

Sol^{n}: distance covered (s) = 50 m

velocity of the car (v) = 36 $kmh^{-1} = \frac{36 \times 1000}{60 \times 60} = 10\ ms^{-2}$

mass of the car (with passenger) (m) = 2000 kg

Now, $v^2 = u^2 - 2as \qquad \Rightarrow 0 = 10^2 - 2 \times a \times 50 \qquad \therefore a = 1\ ms^{-2}$

$F = m \times a = 2000 \times 1 = 2000$ N.

Again, if t is the required time, then

$v = u + at \Rightarrow 0 = 10 - 1 \times t \qquad \therefore t = 10$ secs..

9. A force equal to a weight of 1 kg acts on a body continuously for 10 secs and causes it to distance 10 metres in that time, find the mass of the body.

Sol^{n}: force (F) = wt. of 1 kg = $1 \times 9.8 = 9.8$ N

Initial velocity (u) = 0 m/s

$\therefore \quad$ acceleration of object (a) = ?

Now, $s = ut + \frac{1}{2}at^2$

$\Rightarrow \quad 10 = \frac{1}{2} \times a \times (10)^2$

$\Rightarrow \quad a = 0.2\ m/s^2$

Again, F = ma $\qquad \Rightarrow 9.8 = m \times 0.2 \qquad \therefore m = 49$ kg.

10. Due to the application of a force of 12 kg.wt. a body of mass 5 kg changes its velocity from 12 m/s to 20 m/s. Find the distance through which the body described.

Sol^n: mass (m) = 5 kg

Force (F) = 12 kg wt = 12×10 N = 120 N

Now, $F = ma \Rightarrow 120 = 5 \times a \Rightarrow a = 24$ m/s^2

We have, $v^2 = u^2 + 2as$

$\Rightarrow 20^2 = 12^2 + 2.24.s \Rightarrow s = 5\frac{1}{3}$ m.

11. Find the velocity of a 4 kg shot that will just penetrate through a wall 25 cm thick, the resistance being 36 tones wt. ($g = 9.8$ ms^{-2})

Sol^n: mass of shot (m) = 4 kg

penetrated space (s) = 25 cm = $\frac{1}{4}$ m

resistance (F) = 36 tones wt = $36 \times 1000 \times 9.8$ N

final velocity (v) = 0 m/s

initial velocity (u) = ?

If a is the retardation produced by the wall

Then, $F = -ma \Rightarrow a = -\frac{F}{m} = \frac{36 \times 1000 \times 9.8}{4} = -88200$ m/sec^2

Again, $v^2 = u^2 + 2as$

$\Rightarrow 0^2 = u^2 - 2 \times 88200 \times \frac{1}{4} \Rightarrow u^2 = 44100 \therefore u = \sqrt{44100}$ ms^{-1} = 210 ms^{-1}.

12. A body of mass 1 kg is falling under gravity at the rate of 28 ms^{-1}. What is the uniform force that will stop it;

i) 0.1 second, ii) 20 cm ($g = 10$ ms^{-2})

Instead of falling under gravity if the body is moving at the rate of 20 ms^{-1} along a horizontal line, what will be the force required in above two cases.

Sol^n: mass of the body (m) = 1 kg

initial velocity (u) = 28 ms^{-1}

final velocity (v) = 0

The forces acting on the body are

(a) weight of the body, mg = 1 g in downward direction

(b) the force F acting vertically upward direction

Case (i) let a be the retardation after striking the ground before coming to the rest for 0.1 s.

Then, $v = u - at \Rightarrow 0 = 28 - a \times 0.1 \therefore a = \frac{28}{0.1} = 280$ m/sec^2

Then by second law of motion we have

$F - mg = ma$

$\Rightarrow F = m(g + a) = 1 \times (10 + 280) = (10 + 280)$ N = 290 N = $\frac{290}{10}$ kg wt = 29 kg wt.

Case (ii) let a' be the retardation after striking the ground before coming to the rest at the distance of 20 cm = 0.2 m

Then, $v^2 = u^2 - 2a's$

$\Rightarrow 0^2 = 28^2 - 2 \times a' \times 0.2 \Rightarrow 0.4a' = 784 \therefore a' = \frac{784}{0.4} = 1960$ ms^{-2}

Then by second law of motion we have

$F - mg = ma'$

$\Rightarrow \quad F = m(g + a') = 1(10 + 1960)\ N = 1970\ N = \frac{1970}{10}$ kg wt. = 197 kg wt..

Instead of falling under gravity if the body is moving along the horizontal at the rate of $28\ ms^{-1}$, the required force are

i) $F = ma = 1 \times 280\ N = 280\ N = \frac{280}{10}$ kg wt = 28 kg wt. and

ii) $F = ma' = 1 \times 1960\ N = 1960\ N = \frac{1960}{10}$ kg wt = 196 kg wt.

13. A mass of 5 kg falls 300 cm from rest and is then brought to rest by penetrating 30 cm into some sand; find the average thrust of the sand on it.

Sol^n: mass (m) = 5 kg

initial velocity (u) = 0 m/s

height (h) = 300 cm = 0.3 m

Suppose v is the velocity of the body when it falls 300 cm from rest under gravity.

Then, $v^2 = u^2 + 2gh \quad \Rightarrow v^2 = 0^2 + 2 \cdot g \cdot 3 \quad \Rightarrow v^2 = 6g$ ..... (i)

The velocity given by (i) is reduced to zero when the body goes to 30 cm = 0.3 m into sand. If a is the retardation of the system them

$0^2 = v^2 - 2 \times a \times 0.3$

$\therefore \quad a = \frac{v^2}{0.6} = \frac{6g}{0.6} = 10\ g\ m/sec^2$

Let, F be the average thrust of the sand on the body.

Then the forces acting on the body are

(a) The weight 5 g N of the body acting downward.

(b) A force F N of the sand acting upward

Then applying Newton's second law of motion we have

$F - mg = ma \quad \Rightarrow F - 5g = 5 \times 10\ g \quad \Rightarrow F = 55g \quad \therefore F = 55$ kg wt..

14. A particle of mass 15 kg falls from a height of 18 metres and penetrates into some sand. If the average resistance offered by the sand is equal to a force of 150 kg.wt., find how far it penetrates into the sand.

Sol^n: mass(m) = 15 kg

height (h) = 18 m

let v be the velocity just before penetrating, then

$v^2 - u^2 = 2gh \quad [\because u = 0]$

$\Rightarrow \quad v^2 = 2 \times 10 \times 18 = 360 \quad \therefore v^2 = 360$

After striking the sand the forces acing on the body are

i) The weight mg of the stone acting vertically downwards.

ii) The resistance F offered by the sand acting vertically upward.

Then applying Newton's second law;

$F - mg = ma'$

$\Rightarrow \quad F = m(g + a')$

$\Rightarrow \quad 150 \times 10 = 15\,(10 + a')$

$\Rightarrow \quad a' = 90$

Now, retardation of particle (a') = 90 while penetrating the sand s m far from the surface of sand.

Thus we have,

$v^2 = u^2 - 2a's$

$\Rightarrow \quad 0 = 360 - 2 \times 90 \times s \quad [v^2 = \text{final velocity} = 0]$

$\therefore \quad s = 2m.$

15. A mass 'm' kg is acted on by a constant force 'P' kg wt and in 't' secs; it moves a distance of x metres from rest and acquires a velocity of v m/s. Show that $x = \frac{gt^2P}{2m} = \frac{v^2m}{2gP}$.

Sol^n: mass of the body = m kg

force acting = P kg wt. = Pg N

time taken = t secs.

distance described = x metres

initial velocity = 0

final velocity = v ms^{-1}

If a is the acceleration produced, then by second law of motion we have

Force = mass × acceleration produced

$\Rightarrow \quad Pg = ma \quad \Rightarrow a = \frac{Pg}{m}$

Again, $s = ut + \frac{1}{2}at^2$

$\Rightarrow \quad x = 0 \times t + \frac{1}{2}\frac{Pg}{m}t^2 \qquad \Rightarrow x = \frac{gt^2P}{2m}$ ..... (i)

Also, $v^2 = u^2 + 2ax$

$\Rightarrow \quad v^2 = 0^2 + 2 \cdot \frac{Pg}{m} \cdot x$

$\Rightarrow \quad v^2 = \frac{2Pgx}{m} \qquad \Rightarrow x = \frac{v^2m}{2gP}$ ..... (ii)

Combining (i) and (ii), we get

$x = \frac{gt^2P}{2m} = \frac{v^2m}{2gP}$

16. A balloon is rising with an acceleration f. Prove that the fraction of the weight of the balloon which must be emptied out of the balloon in order to double the acceleration is $\frac{f}{g + 2f}$.

Sol^n: If the mass of the balloon is m, then its weight = mg

Let, f be the acceleration, F be the upward lifting force of the balloon. Then, by Newton's second law of motion we have,

$F - mg = mf \qquad$ or, $F = mg + mf$ ..... (i)

Suppose, x be the mass to be taken out from the balloon so that it may now move up with acceleration 2f

Then $F - (m - x)g = (m - x)2f$

or, $mg + mf - mg + xg = 2mf - 2xf$

or, $xg + 2xf = 2mf - mf$

or, $(g + 2f)\, x = mf$

$\therefore \quad \frac{x}{m} = \frac{f}{g + 2f}$